📄 Product Requirements Document (PRD)

🩺 Product Name: MedAI

Version: 1.0
Product Type: AI-Powered Health Information Assistant
Platform: Web Application

1️⃣ Product Overview

MedAI is a web-based AI health assistant that allows users to:

- Get detailed information about medicines
- Enter symptoms to receive possible condition insights
- Maintain personal health interaction history
- Access the platform securely via Google login

⚠️ MedAI is not a diagnostic tool. It provides informational guidance only and encourages users to consult licensed medical professionals.

2️⃣ Problem Statement

Many people:

- Google medicines and get confusing information
- Self-diagnose from unreliable sources
- Panic after reading medical forums

- Don’t know correct dosage guidance

There is a need for:

✔ Clear
✔ Structured
✔ Calm
✔ Trustworthy
✔ AI-simplified health information

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3 Goals & Objectives

Primary Goals

- Provide accurate, structured medicine information
- Offer AI-based symptom analysis (non-diagnostic)
- Create a safe and trustworthy user experience
- Build a scalable AI-integrated medical information system

Success Metrics

- User signups
- Daily active users
- Average session time
- Number of symptom checks
- User retention rate

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4 Target Audience

- Students

- Working professionals
- Parents
- Elderly users (simple UI required)
- Users searching medicine information
- People wanting preliminary symptom understanding

5️⃣ Core Features (MVP)

🔐 5.1 Authentication

- Google Sign-in
- Secure session handling
- User profile storage
- Logout functionality

💊 5.2 Medicine Information Module

User can:

- Search by medicine name
- Ask natural language questions

System provides:

- What the medicine is used for
- How it works
- Typical dosage (general guidance only)

- Side effects
- Warnings
- Drug interactions (basic)

Sources:

- AI model
- OpenFDA API (optional enhancement)

🩺 5.3 Symptom Checker Module

User can:

- Enter symptoms in natural language
- Select from symptom checklist (optional enhancement)

System provides:

- Possible conditions (not diagnosis)
- Why those conditions match
- Recommended general actions
- Clear disclaimer

Response format must be structured:

- Possible Condition
- Reason
- Risk Level (Low/Moderate/High)
- Suggested Next Step

💬 5.4 Chat-Based AI Interface

- Conversational UI
- Chat history stored per user
- Timestamped logs
- Ability to clear history

📜 5.5 Medical Disclaimer System

Display:

- On homepage
- Before first symptom check
- On every AI response footer

Example:

> “This tool provides informational guidance only and does not replace professional medical advice.”

6️⃣ Non-Functional Requirements

🔒 Security

- JWT-based authentication (if not Firebase)
- HTTPS required
- Secure API key storage (dotenv)
- Input validation & sanitization

⚡ Performance

- AI response under 5 seconds
- Optimized database queries
- Scalable backend architecture

📱 Responsiveness

- Mobile-first design
- Tablet and desktop compatible

🎨 UI/UX Requirements

- Calm color palette (white, light blue, soft green)
- Rounded UI elements
- Accessible typography
- Clear spacing
- No aggressive red tones

7️⃣ Technical Architecture

Frontend

- React (Vite)
- Tailwind CSS
- ShadCN UI (optional)
- Axios
- React Router
- Firebase Authentication

Backend

- Node.js
- Express.js
- MongoDB (Mongoose)
- OpenAI API
- OpenFDA API (optional)

Deployment

- Frontend: Vercel
- Backend: Render / Railway
- Database: MongoDB Atlas

8️⃣ Database Schema (High Level)

User

- _id
- name
- email
- googleId
- createdAt

ChatHistory

- userId
- message

- response
- type (medicine / symptom)
- timestamp

9️⃣ AI System Requirements

AI must:

- Avoid definitive diagnosis
- Avoid emergency medical instructions
- Avoid prescribing restricted medicines
- Provide structured response format
- Include disclaimer automatically

System prompt must enforce safety boundaries.

🔟 Legal & Compliance

- Privacy Policy page
- Terms & Conditions page
- Cookie consent (optional)
- No medical claim marketing
- Label product as “AI Health Information Assistant”

1️⃣1️⃣ Future Enhancements (Phase 2)

- Medicine reminder system
- Save favorite medicines
- PDF health report export
- Voice input
- Multi-language support
- Dark mode
- Admin dashboard with analytics

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12 Risks & Mitigation

Risk	Mitigation
Incorrect AI output	Strict system prompt
User misuse for diagnosis	Clear disclaimers
Sensitive data storage	Encrypted database
Legal concerns	Avoid medical claims

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13 MVP Timeline (Suggested)

Week 1:

- Authentication
- UI skeleton

Week 2:

- Backend API
- AI integration

Week 3:

- Symptom checker
- Medicine module

Week 4:

- Testing
- Deployment
- Documentation

1️⃣4️⃣ Unique Value Proposition

Unlike random search results, MedAI provides:

- Structured responses
- Calm and readable format
- AI-personalized guidance
- Secure user accounts
- Clean medical-focused interface

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